DO NOT WRITE IN THIS AREA

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**16** Here are the first five terms of a quadratic sequence.

10 21 38 61 90

Find an expression, in terms of $n$, for the $n$th term of this sequence.

First differences: 11, 17, 23, 29; second differences: 6.

So quadratic term is $3n^2$. Subtract it: 7, 9, 11, 13, 15.

Remainder is $2n + 5$; nth term = $3n^2 + 2n + 5$.

**3 marks total**
Second diff 6 gives $3n^2$
Linear remainder 2n+5
$3n^2+2n+5$

**(Total for Question 16 is 3 marks)**